

UNIT - 2

TEACHING SKILLS

2.1 Micro Teaching

A Teacher make use of a number of methods and techniques to make the teaching effective, successful and interesting. The technique includes motivating, explaining, questioning, writing on the black board, using teaching aids and so on. The teacher could also make use of non verbal behaviours such as smiling, nodding and gesturing. These group of activities are called skills. So teaching is a various combination of skills.

Microteaching introduces to the teacher trainee to a wide range of teaching skills and allows the teacher trainee to practice each skill one at a time until he or she becomes proficient in the skill.

2.1.1 Microteaching - Definition

The word Microteaching was formed by A.W.Dwight Allen during 1963 defined microteaching "as a scaled down teaching encounter in class size and class time".

Allen Eve defined microteaching "as a system of controlled practice, that makes it possible to concentrate on specific teaching behaviour and to practice teaching under controlled conditions."

Microteaching is reduced teaching. Here new skills are polished - M.C. Knight 1971.

Microteaching is a teacher training technique. Here a teacher trainee teachers a short lesson of about five minutes duration to a small group of students practicing one teaching skill at a time.

2.1.2 Need and importance of Microteaching

In teaching learning process, we can't use single skill. At the same time if a teacher wants to be good in teaching, he should be given training in all teaching skills one by one. For this micro teaching is more useful.

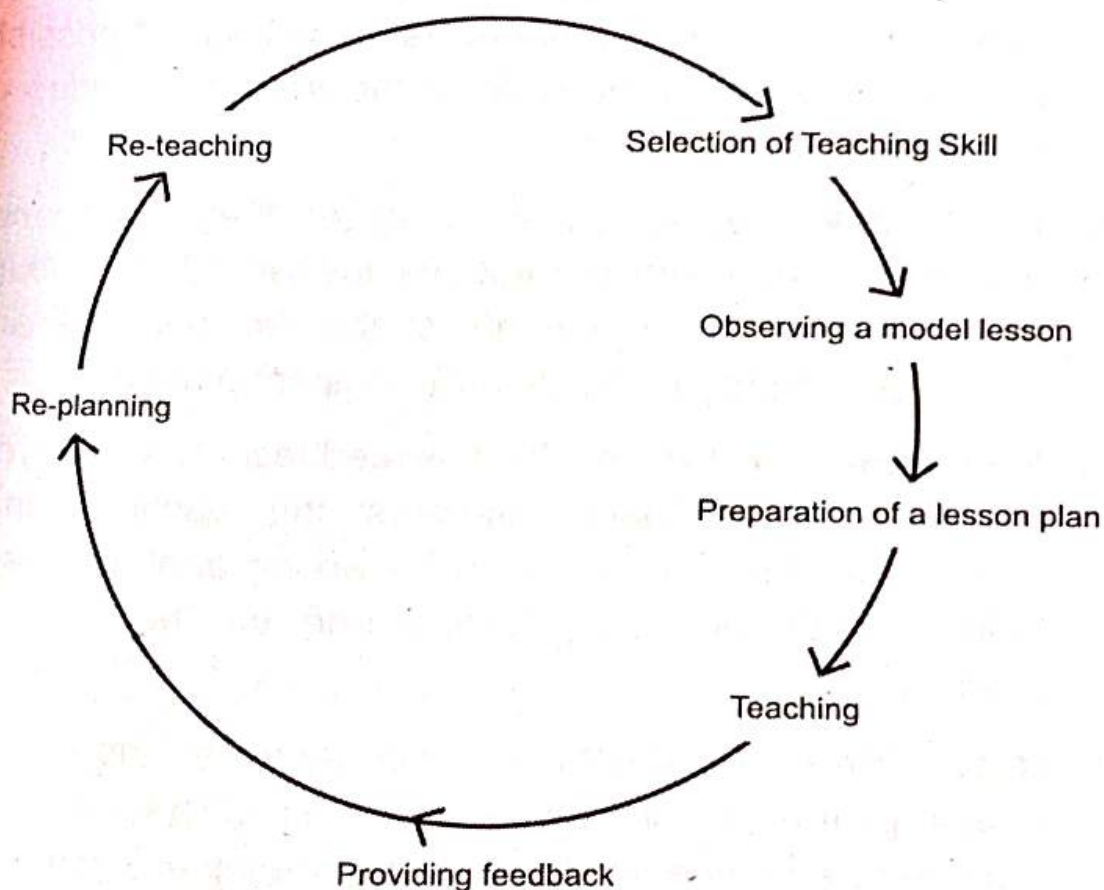
Here each skill is looked liked through microscope. Instead of "What to teach", how to teach is considered. The beauty of this method is, teacher trainee get unproblematic artificial atmosphere instead of natural class room atmosphere. This method is very helpful to improve the teaching skills and to remove the difficulties in teaching.

2.1.3 Characteristics of microteaching

- Microteaching is a teacher teaching technique and not a teaching method.
- It is a real teaching but if focuses on developing teaching skill and competencies.
- In microteaching the teacher trainee practices one specific teaching skill at a time, till he/she attains mastery over the skill.
- Class size is reduced to 5 - 6 students.
- Duration of the class is 5 - 6 minutes
- Microteaching is a cyclic process, the predecided model is plan, teach, feedback, re-plan, re-feedback. It is repeated till the trainee achieves the expected level of mastery.
- Microteaching is practiced in terms of definable, observable, measurable and controllable teaching skills.

2.1.4 Microteaching Cycle

Microteaching is a method that enables a teacher - trainee to practice one teaching skill till he or she attains mastery over the skill. The practice phase is the most important phase of the microteaching programme. In this phase the teacher trainee chooses a specific skill, prepares a micro-lesson plan and teaches a small group of students for a duration of 5 - 6 students. We follow few steps to get training in the skill. These steps are called microteaching cycle.



Steps in microteaching cycle

1. **Selecting a skill** : This involves selection of a teaching skill practiced, knowing its components clearly.

2. **Observing model lesson** : Teacher educator will present a model microteaching lesson plan of the selected skill. Trainee observe it carefully
3. **Preparing Lesson Plan** : Teacher trainee has to prepare a micro-lesson plan for 5 - 6 minutes for the selected teaching skill. In lesson plan all the components of the skill should be presented in an appropriate place.
4. **Teaching** : The trainee teaches the lesson for 5-6 minutes. There will be 5-6 students or peers with teacher educator or one or two peers as supervisor. If possible record the teaching methods of the trainee by video or audio tape.
5. **Feed back**: The supervisor analyses the performance and discusses it with the teacher trainee. The feedback should focus on components of the teaching skill and draw attention to modification for mastering the skill.
6. **Replanning**: In the light of the feedback received from the supervisor or peer observers, the teacher trainee replans her micro-lesson plan by writing another lesson plan or modifying the existing one on the basis of feedback.
7. **Reteaching**: The teacher-trainee reteaches the revised lesson plan to another group of students. The supervisor checks to see whether there is any improvement in skill attainment. This process repeats till the teacher trainee acquires the required level of competency.

2.1.5 Uses of microteaching

Microteaching techniques increases the effectiveness of the teacher training programmes in the following ways

1. Microteaching helps in reducing the complexities of the normal classroom teaching. This helps the teacher trainees gain more confidence in classroom teaching.
2. Microteaching gives the knowledge of various teaching skills.
3. It is useful for developing teaching efficiency.
4. Microteaching prepares trainee to effective and better teacher.
5. Microteaching creates among the teacher trainees an awareness of the various skills of which teaching is composed of.
6. Each trainee makes progress in developing teaching skills at his own speed depending his ability.
7. Using video or Audio tape enable the trainee to analyse his own teaching performance.
8. Feedback enable the teacher-trainee to consciously concentrate on specific behaviour modification.\

2.1.6 Micro Teaching Vs Macro Teaching

Sl.No.	Micro Teaching	Macro Teaching
1.	It is a teacher training technique.	It is a method of classroom teaching
2.	Class size is 5 to 10 students	Class size is more than 40
3.	Duration of the class 6-10 minutes	Duration of the class 40 to 45 minutes
4.	One or two teaching Skills used	Maximum Teaching skills are used.

Sl.No.	Micro Teaching	Macro Teaching
5.	Importance is given to skill	Importance is given to content
6.	It puts the teacher under the microscope	No such arrangement for macro teaching
7.	All the faults of the teacher are observed	No such arrangement for macro teaching
8.	Feed back is given	No feed back is given
9.	Opportunity to reteach the lesson	No such arrangement
10.	No importance is given to students needs	Importance is given to students needs
11.	No classroom activities for students	Classroom activities will be given.
12.	It tends to reduce the creativity of the teachers	It increases the creativity of the teachers
13.	Micro teaching can be successful in a controlled environment only	Macro teaching can be successful in any environment
14.	Teaching is relatively simple	Teaching becomes complex
15.	Micro teaching is done with all sensory resources	Macro teaching relies on auditory learning and recall.

2.2 Identification of teaching skills:

A teaching skill is a set of activities or actions of the teacher which are repeated a number of times during the course of teaching.

Characteristics Of Teaching Skills

- a. Teaching skills are the specific instructional activities.
- b. Teaching skills are verbal and non-verbal in nature.
- c. A teaching skill includes activities and actions.
- d. Teaching skills involve the teacher's activities which are performed in classroom.
- e. Teaching skills develop the communicative competency.
- f. Teaching skills help effective presentation of content in classroom.

List of Teaching skills given by Allen and Ryan

1. Stimulus variation
2. Set induction
3. Closure
4. Teacher silence and non-verbal cues
5. Reinforcing pupil participation
6. Fluency in questioning
7. Probing questioning
8. Use of higher order questions
9. Divergent questions
10. Recognising and attending behaviour
11. Illustrating and use of examples
12. Lecturing

13. Planned repetition

14. Completeness of communication

List of Teaching skills given by B.K. Passi

1. Writing instructional objectives
2. Introducing a lesson
3. Fluency in questioning
4. Probing questioning
5. Explaining
6. Illustrating with examples
7. Stimulus variation
8. Silence and non-verbal cues
9. Reinforcement
10. Increasing pupil participation
11. Using blackboard
12. Achieving closure
13. Recognizing attending behaviour

Types of Teaching skills selected

Large number of teaching skills have identified in bringing effective teaching. Allen and Ryan listed fourteen teaching skills and in 1983 Menon gave a list of seventy four teaching skills. A few important skills listed below will be discussed here.

1. Skill of Set induction
2. Skill of Explaining
3. Skill of blackboard usage
4. Skill of questioning

5. Skill of probing questions
6. Skill of stimulus variation
7. Skill of Reinforcement
8. Skill of Non-verbal clues
9. Skill of closure

2.2.1. Skill of Introducing the Lesson / Skill of set induction

In teaching, the way of introducing a lesson is more important. The proverb "Good beginning is half of the work completed." is also fit for teaching. The teacher, before introducing a lesson, has to motivate the students and establish cognitive and affective rapport with the students. The following are the components of skill of introducing the lesson.

1. Testing previous knowledge
 - a. Preliminary attention gaining
 - b. Use of previous knowledge
 - c. Use of appropriate device
 - d. Link with a new topic
2. Attention Gaining
 - a. Use of voice to focus attention
 - b. Use of audio visual aids
 - c. Use of gesture / eye contact
 - d. Introducing something unusual
3. Motivation stimulation
 - a. To arouse curiosity
 - b. Method of introducing a lesson

c. Able to involve students

1. Testing previous knowledge

1a. Preliminary attention gaining

Tell the significance of the topic. Give importance of new knowledge in our life situations.

1b. Use of previous knowledge

Before giving new knowledge, teacher has to test the previous knowledge which is to be used in the new topic.

1c. Use of appropriate device

By way of asking few questions or asking students to write two or three sentences about the new topic.

1d. Link with a new topic

Integrate the old knowledge with the new knowledge. Use the knowledge obtain from Text book, Library book etc., to teach the new topic.

2. Attention Gaining

2a. Use of voice to focus attention

Use clear and bold voice to focus attention.

2b. Use of audio visual aids

Use appropriate teaching aids required to teach the topic to arouse pupil's attention. Charts, slides and power point presentation may be used.

2c. Use of gestures / eye contact

Gestures are indication of conveying messages by body movements. Eye contact is a signal for lack of interest in a person. Much of eye contact shows that it is a threatening attitude.

2d. Introducing something unusual

Relevant to the new lesson / topic, show a chart, photograph or a instrument which is not seen by student,

3. Motivation stimulation**3a. To arouse curiosity**

To bring in interest on the topic the teacher must refresh the students properly. He / She should enact to the situation. Must quote a lot of day to day happenings of the new knowledge and give a lot of examples. Self involvement is needed to bring in interest among students. He must become a part of lesson, only then he can arouse curiosity in the students.

3b. Method of introducing a lesson

Special care should be given to introduce a lesson. Teacher has to select any method / technique to introduce a lesson from the methods given below according to the nature of the lesson.

1. Telling relevant statements to teach the new topic.
2. Introduce the lesson after obtaining answers from the students.
3. By way of anecdotes, Explaining or commenting, introduce a lesson.
4. By story telling technique, introduce a lesson.
5. By the method of dramatisation, introduce a lesson.
6. Using teaching aids, we introduce a lesson.
7. By the method of demonstration, introduce a lesson.

3c. Able to involve students

While introducing a lesson teacher can ask students to collect or prepare materials / Aids well in advance.

2.2.2. Skill of Explaining

For an effective learning, explanation in teaching should be done effectively. Explaining bridges the gap in understanding the new knowledge by relating it to the past knowledge. Thus explaining depends upon the past knowledge, the type of new knowledge and the relationship among them. Explanation can be more effective by using simple and clear language for clarity. The following are components of the skill of explanation.

1. Beginning statement
2. Explaining links
3. Concluding statement
4. Questions to test pupil's understanding
5. Irrelevant statements
6. Lacking in continuity
7. Vague words and phrases

1. **Beginning statement:**

While explaining the beginning statement must include central concept of the content. Statements such as 'Today we are going to derive a formula or prove a theorem' etc. Beginning statement must gain attention of the students.

2. **Explaining links:**

While explaining the subject matter use suitable link words between the concepts. examples: therefore; that is why, since, because, hence, in order to etc.

3. **Concluding statement:**

Concluding statement should summarise what the teacher explained. Students can summarise what they learn if concluding statements are meaningful.

4. Questions to test pupil's understanding

Questions asked here are from the content explained. Teacher check the understanding of students by questioning. These questions will also help to analyse the performance of the explanation given by the teacher.

5. Irrelevant Statement

Teachers should avoid irrelevant statements while explaining the topic.

6. Lacking in continuity

Knowledge given in classroom are information, facts, concepts etc. It should be given in sequence and in continuity. Then only the students clearly understands the concepts, facts and information.

7. Vague words and phrases

The teacher should not use vague words and phrases which is not understandable by students. The vocabulary used should be clear and according to the level of students.

2.2.3. Skill of Blackboard usage

Among ten major skills of micro-teaching, the skill of blackboard writing is one of them. Blackboard is the visual aid that is widely used in all sectors of education and training. It is used as a powerful teaching aid for classroom instruction in each and every level of education. Black board is the most suitable tool for providing a holistic picture of the lesson. Blackboards can be suitably used for displaying notes and diagrams during a less and for working through calculations providing theorems, constructing geometrical figures or similar exercises. Thus if the blackboard is used properly, it adds variety to the lesson and draws the attention of the pupil.

Components of the skill of Blackboard usage

The six components of the skill of Black Board usage are

1. Legibility
2. Comprehension
3. Highlighting the main points
4. Compatibility
5. Blackboard summary
6. Contact with the pupil

1. Legibility

The handwriting of the teacher on the black board should be legible so that it can draw the attention of the learners and encourages them to improve their own handwriting. The size of the capital letters should be larger than that of the small letters and the handwriting should be as vertical as possible without deviation from the line. While cleaning the black board rub it from Top to bottom.

2. Comprehension

While writing on the black board, the teacher should be conscious enough of the fact that the size of the letters written on the blackboard should be uniform and large enough to be read by the students in the last row. Overwriting the letters should be avoided. Numbers and letters should be written clearly. Grammatically correct sentences should be written on the black board.

3. Highlight the Main points

The main points or words are written on the black board should be highlighted by underlying them. Colored chalk should be used suitable for the purpose of learner's attention. However too much use of coloured chalk should be avoided.

4. Compatibility

There should be a clear distinction between every letter as well as adequate space should be maintained between individual letters and words. Important words or statements should be written with proper space. Take care that only essential materials should be written on the blackboard.

5. Blackboard Summary

The teacher should take much care and attention to develop a black board summary at the end of the lesson. It makes teaching meaningful and interesting. The black board summary should be so brief that the students can recollect the whole lesson at a glance.

6. Contact with the pupils

The teacher at the time of writing on the blackboard should maintain eye contact with his students. This is necessary for controlling interactions, maintaining, discipline, sustaining the attention of the learners.

2.2.4 Skill of questioning

Successful teaching technique is highly depend on questioning. It is a skill that can be used in teaching any subject and any standard. In a person different types of questions, depending upon situation and purpose to be achieved are used. Questions are used to help students recall certain facts, exercise their reasoning ability, and use their power of recognition and discrimination in enhancing their learning. Effective questioning encourages them to reflect over things and ideas and leads them to participate in discussion.

Purpose of questioning

- Questioning is a logical procedure of problem solving

- Questioning is useful for presenting the tentative nature of knowledge.
- The teacher encourages the learners to seek more than one answer for a question
- The teacher uses questioning to achieve learning objectives
- The teacher uses questioning to assess students understanding.

Thus questioning promotes involvement, enhance learning and motivate students.

Let us examine why questioning is an essential skill for a teacher.

- It helps students test their knowledge
- It explicit desired information from students
- It stimulates thinking
- It develops subject matter
- It enable students to analyse the problem
- It enable students to apply their knowledge to a specific new situation.
- It helps students evaluate for themselves their understanding of the subject matter.
- It motivate students to participate in the teaching-learning process.

Questioning stimulates thinking. The level of students thinking depends upon the level of questions asked by the teacher. The level of question is also determined by the level of the responses given by the students. If order to elicit responses at different levels from the students, we can frame questions at different levels.

Questioning is one of the most important skill in teaching. It has to be developed by any teacher. While teacher teaches a lesson he has to ask several questions related to the content. These type of questions are called teaching questions. After completing a lesson, questions will be asked to know whether the students understood the content of teaching. These questions are called review questions. Teachers should be expert in asking questions. Good questions serve many purposes. Questions stimulate thinking. Questions help recalling facts, Concepts and principles. Questions make active involvement in learning. Questions arose the curiosity of the students

Generally these are three levels of question. They are
1. Low order questions 2. middle level questions 3. higher order questions.

1. Low order questions.

These questions require the students to recall knowledge that they have learnt. The questions of this level do not ensure understanding of the content or analyzing information, but requires memorization.

Examples : What is the area of the circle?

State pythagoros theorem.

Write the expansion of $(a+b)^2$.

2. Intermediate level of questions.

Explaining the meaning of a concept or statement, compare or differentiate among two things etc are intermediate level of questions.

1. Find any two rational number between $\frac{4}{5}$, $\frac{6}{7}$
2. Find the cube root of $\frac{125}{216}$

3. Surface area of a hemisphere is 1232 cu.cm. Find its diameter.

3. Higher order questions

These type of questions requires the students to use higher levels of intellectual abilities such as comprehension, application, analysis, synthesis and evaluation.

Example:

1. Draw a quadrilateral ABCD using the given measurements and find its area $\overline{AB} = 5.5$ cm, $\overline{BC} = 4.5$ cm, $\overline{AC} = 6.5$ cm, $\overline{AD} = 4.2$ cm, $m\angle CAD = 80^\circ$
2. What are the conditions that triangle $\triangle ABC$ and $\triangle PQR$ are congruent?

2.2.5 Skill of probing

The skill of probing questions involves going deal into pupil's response by asking a series of subsequent questions in a topic. It helps the pupils to give a correct and complete answer. This skill requires the teacher to put a series of questions about the answer given to the first question. It is a major micro-teaching skill.

When students are unable to give the answer or give incomplete answers, teachers use probing questions.

Purpose of probing

1. To motivate the students for correct answer
2. Ask many questions related to the main subject
3. Taking various supportive questions
4. Linking previous knowledge with the new knowledge
5. Creating interest and curiosity among students.
6. Developing the thinking, creativity and analytical ability

Components of probing skill of questioning

There are five main components of skill of probing

1. Prompting
2. Seeking further information
3. Re-Focusing
4. Re-Direction
5. Critical awareness

1. Prompting

Prompting mostly involves verbal questions, visual questions and some visual guidance. In order to help the student to know the correct answer, the teacher gives some indication.

2. Seeking further information

Most of the time student's don't give the proper and accurate answer. Then the teacher asks the same question to other students in order to get the correct answer. The teacher asks different aspects of the question in order to seek further information. This also may check the students' knowledge.

3. Re-Focusing

If the teacher is not satisfied with the answer of the student then the teacher puts different situations in front of the students which helps them in the transfer of knowledge of the students.

4. Re-Direction

The teacher tries to know the multiple answer to the question in order to make students think again and again. This will increase the thinking ability of the students, and this

may increase the participation of the students in the class too.

5. Critical awareness

If the teacher asks why? how? type of questions then it is called Critical Awareness. This critical awareness among students increase the rational thinking and awareness.

2.2.6. Skill of stimulus variation

While teaching teacher must draw attention of the students. For this, teacher uses some gestures, body movements, makes certain verbal statements, modulating the voice etc. All these behaviours are related to stimulus variation. Components of this skill are :

1. Movements
2. Gestures
3. Change in speech pattern
4. Change in inter action style
5. Secure attention
6. Pausing
7. Oral - Visual switching

1. Movements

Teacher should not stand in one place while teaching. He has to move from one place to another to write on the blackboard, to go near and appreciate a student, to move towards a model etc. Hence, in order to gain students attention a teacher has to move purposeful movements.

2. Gestures

Use of appropriate gestures according to the content to emphasis importance, to explain emotions, make students to understand the message. Hand movement to describe

shape, size, Body movements and facial expression enhance the value of message.

3. Change in speech pattern.

If a teacher explains the content in the same modulation, then it is boring. Hence, in teaching there should be voice modulation to pay attention to a particular point. Modulating the voice on situations, change in pitch, speed of voice will force his student to pay attention.

4. Change in interaction style

Teacher has to plan, how students also involve in teaching. Asking questions from the students, asking student to do a sum on the blackboard, students are encouraged to put few questions and teacher will answer it, to participate students in a seminar or discussion and teacher will tell its result etc. are all called change in interaction. This can be mention in a simple form as

Teacher - Class

Teacher - Student

Student - Student

5. Secure Attention.

While teaching the teacher may use certain statements like, this is important, look at this diagram, listen carefully etc to draw attention. While writing on the blackboard teacher draw attention by using colour chalk for important message or underlined the important points.

6. Pausing

Pausing means introducing silence during teaching. From this act, we can draw attention of the students. While teaching continuously or at the time of asking questions teacher has to plan pausing and create attention.

7. Oral - Visual switching

While the teacher orally explains, student's hearing. When the teacher exhibits aid or writes on blackboard his visual sense is used. Such frequent changes from oral to visual or from visual to oral help to secure pupil's attention.

2.2.7 Skill of Reinforcement / Skill of Appreciation

All these behaviours are related to stimulus variation. The skill of reinforcement refers to the effective use of reinforcers. While teaching, a teacher asks questions and students replies it with correct answer. If the teacher simply goes to another question, then students will not happy. If the teacher reinforce a correct answer with a smile or praise the student, then they will be happy. This is called reinforcement. Teacher has to practice using positive reinforcers and avoid the use of negative reinforcers in the classroom. By Reinforcement, students participation is more in classroom activities. The different components of this skill are

1. Positive verbal reinforcement
2. Positive non verbal reinforcement
3. Negative verbal statement
4. Negative non verbal reinforcement
5. Wrong use of reinforcement
6. Inappropriate use of reinforcement

1. Positive verbal reinforcement

When the students give correct response for a question the teacher has to give positive verbal reinforcement such as Good, Very good, Well, Excellent, Continue etc

2. Positive nonverbal reinforcement

When a student give correct response for a question the teacher may give non-verbal reinforcement expressing happiness or with the help of body language. Teachers should use more positive reinforces so that the students participation is maximum in teaching activities.

3. Negative verbal statement

Teacher should avoid using these type of negative verbal reinforces namely, I do not like you, stop it, nonsense, you won't improve, you will fail in the examination etc.

4. Negative non verbal reinforcement

Teacher should not use these type of Non-verbal negative reinforces such as facial expression like, frowning, sadness, moving away from responding student, looking at angrily etc.

5. Wrong use of reinforcement

When a student gives a wrong answer, teachers should avoid using words like idiot, stop, enough or showing terror face etc. which will hurt student and his enthusiasm will decrease in the subject.

6. Inappropriate use of reinforcement

Teacher should not use reinforces in a delayed manner or using it always. Using the above type of reinforcement affects the learning and decrease the motivation of the students.

2.2.8. Skill of Non-Verbal cues

Non-verbal cues has been defined as communication without words. Although verbal output can be turned off, non verbal can't. Teachers use Non verbal cues because

1. Words have limitations. There are numerous areas where non verbal communication is more effective than verbal.
2. Non verbal signs are powerful. Non verbal cues primarily express inner feelings.
3. Non verbal messages are likely to be more genuine.
4. Non verbal signals can express feelings, thoughts etc.
5. Complexity of the verbal message can be given through simple non verbal signals.

Components of the skill

1. Body language
2. Proximity
3. Touch
4. Eye contact
5. Chronemics
6. Smell
7. Vocalics

1. **Body language** : Body movement have specific verbal transactions. To explain the subject matter or to appreciate the students we use body language. To describe shape, size body movements are helpful.
2. **Proximity**: When a teacher wants to convey message to a student or to control a problematic behaviour student he has to go near him and convey non verbally.
3. **Touch** : Physical contact with others is the most basic form of communication and certain situations often communicates that there is a problem.

4. **Eye contact :** It is a signal for lack of interest in a person. Too much eye contact shows that it is a threatening attitude.
5. **Chronemics:** With the help of chronemics we can draw attention of the students by means of waiting and pausing. While teaching teacher has to plan waiting and pausing. By increasing the waiting time, you buy for yourself an opportunity to hear and to think. Increase your waiting time upto 5 seconds or longer for slow and shy students and increase in waiting time help students to respondent higher cognitive skills.
6. **Smell :** We can make identify things using smell. This can be used largely in lower standards.
7. **Vocalics :**

Postures : Postures can be used to determine participants degree of attention or involvement. Posture is understand through such indicators as direction of learn body orientation, arm position and body openness.

Adornment : Teacher should wear decent dress. If the school introduce uniform to teachers, teachers should strictly follow them. Lady teachers should not come with more jewels and shining dresses. Clothing should reflect our culture. Hair style should not be funny or attractive.

Locomotion : Teacher should not stand or sit in one place during teaching. He has to move one place to another place or go near to a student to convey non verbal messages.

2.2.9. Skill of closure

After teaching a subject matter in a better way, it is better to close it in the same way. Then only the students feel pleasure and satisfaction. At the end of the lesson, it is necessary to make a quick review. What has been taught,

applying new knowledge in various situations, associating new knowledge with old knowledge and repeating important points. Various components of this skill are:

1. Consolidation of major points.
2. Application of present knowledge.
3. Linking past knowledge with present knowledge.
4. Linking present knowledge with future learning.

1. Consolidation of major points

While teaching teacher has explained so many facts to explain the subject matter. At the end he has to summarise it in a meaningful way. Then only students identify what they have learned.

2. Application of present knowledge

After teaching new knowledge, the teacher has to explain how and where it is used with examples. Then only the students know the importance of new knowledge and its application. For this the teacher may use problem solving method.

3. Linking past knowledge with present knowledge

After explaining the new knowledge, teacher should link it with the past knowledge. Then only the students get a clear view about what they learned and feel that their knowledge has developed.

4. Linking present knowledge with future learning

The new knowledge learned by the students should be linked to the knowledge what they will go to study in future or in continuation to this knowledge. From this they will be ready to learn more about this knowledge.

2.2.10 Advantages or merits of Micro teaching

- It offers opportunity to practice a real lesson without the complexities of a normal classroom.
- It offers several sources of immediate feedback.
- One skill is learnt at a time which is observable, controllable and practicable.
- It helps in the development of self-confidence.
- It helps the trainees to develop the skill of blackboard writing.
- It is helpful to improve his/her own teaching.
- It makes trainees familiar with various teaching styles, eg. Black-board writing, use of aids, asking questions etc.
- Evaluation procedure of a micro-lesson is more precise, objective and acceptable.
- It saves time and energy of the student-teacher as well as the teacher - educator.
- It focusses attention on the modification and improvement of teacher behaviours in the teaching learning process.

2.3 Model Micro Lessons

Model Micro Lesson - 1

Name of the Teacher trainee : xxxxx

Class : Peer Group

Skill : Skill of stimulus variation

Lesson : 8th Std Mathematics

Topic : Geometry - Area of rhombus
 Time : 5 minutes
 Date : xxxx

Components of skill

1. Movement
2. Gestures
3. Change in speech pattern
4. Change in interaction style
5. Focusing
6. Oral visual switching

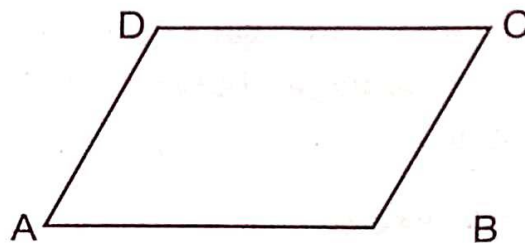
Teacher : Good morning Students

Students : Good morning Sir

Teacher : Today we are going to derive a formula for the area of a rhombus. Teacher moves towards the blackboard and writes the topic (1)

Teacher : Ramu, come and draw the figure of a rhombus on the blackboard (4)

Ramu : Draws the figure and names it ABCD (7)



Teacher : Good (Smiles) (2), Now look at the figure (5) and "tell me whether the rhombus can be divided into two triangles" (3)

Student : By drawing a diagonal we can divide the rhombus into two triangles (4)

Teacher : Teacher nods his head approvingly (2)

Teacher : Rahim, go to blackboard and draw the diagonal (4)

Rahim : Rahim draws the diagonal AC (7)

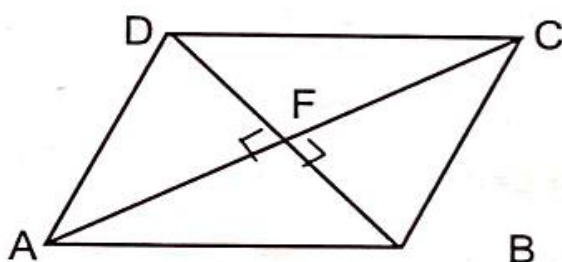
Teacher : Look at the figure (6). Area of rhombus is equal to the area of $\triangle ABC$ and $\triangle ACD$. What is the area of the $\triangle ABC$? (7)

Student : Area of the $\triangle ABC$ is $\frac{1}{2}bh$ (4)

Teacher : Teacher moves toward a student (1) and asked "How you will find h? (3)

Student : By drawing perpendicular (4)

Teacher : Teacher moves (1) towards the board and draws the altitude BF



Teacher : Similarly, how you will find the altitude of $\triangle ACD$? (3)

Student : Drawing an altitude from D to AC (4)

Teacher : Teacher calls a student and asked him to draw the altitude DF (7)

Teacher : Hence area of the rhombus ABCD is (writing on the Black Board) (7)

$$= \frac{1}{2} AC \times BF + \frac{1}{2} AC \times DF$$

$$= \frac{1}{2} AC (BF+DF)$$

$$= \frac{1}{2} AC (BD)$$

Teacher : Look at the figure (5) What is AC (3)

- Student : AC is one diagonal (4)
- Teacher : What is BD?
- Student : BD is another diagonal (4)
- Teacher : So (6) Area of rhombus = $\frac{1}{2} d_1 \times d_2$

Model Micro Lesson - 2

Name of the Teacher trainee : xxxxx

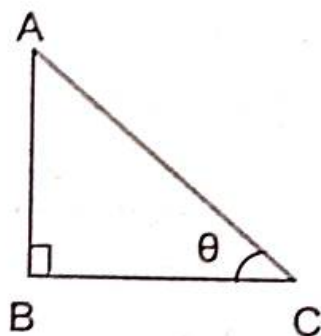
- Class : Peer Group
- Skill : Set Induction / Introducing
- Lesson : 9th Std Mathematics
- Topic : Trigonometry-Trigonometric ratios
- Time : 5 minutes
- Date : xxxx

Components of skill

1. Testing previous knowledge
2. Attention Gaining
3. Motivation Stimulating

- Teacher : Good morning Students
- Students : Good morning Madam
- Teacher : How many sides are there for a triangle (1)
- Student : Three sides
- Teacher : What are the different triangles on the basis of sides? (1)
- Student : Equilateral triangle, Isosceles triangle and scalene (2)
- Teacher : What are the different triangles on the basis of angles? (1)

- Student : Acute triangle, Right angled triangle and obtuse triangle (2)
- Teacher : What do you mean by right angles? (1)
- Student : In a triangle if any one of the angle is 90° then it is called right angled (2)
- Teacher : In a right angled triangle which is the lengthiest side (1)
- Student : Side which is opposite to 90°
- Teacher : What is the name of the side? (1)
- Student : Hypotenuse (2)
- Teacher : Draws a right angled triangle on the black board and marked the angles (3)



- Teacher : What is the angle of $\angle B$ (3)
- Student : 90°
- Teacher : State the Pythagoras theorem (3)
- Student : In a right angled triangle the square of the Hypotenuse is equal to the sum of the squares of the other two sides (2)
- Teacher : What is the angle of $\angle C$ from the figure (3)
- Student : θ
- Teacher : What is the opposite side of θ (3)
- Student : $\overline{AB}$
- Teacher : What is the adjacent side of θ (3)

Student : $\overline{BC}$

Teacher : Students, today we are going to see different trigonometric ratios (3). In a right angled triangle,

$$\text{we define } \sin \theta = \frac{\text{Opposite side of } \theta}{\text{Hypotenuse}}$$

$$\cos \theta = \frac{\text{Adjacent side of } \theta}{\text{Hypotenuse}}$$

$$\text{and } \tan \theta = \frac{\text{Opposite side of } \theta}{\text{Adjacent side of } \theta}$$

Model Micro Lesson - 3

Name of the Teacher trainee : xxxxx

Class : Peer Group

Skill : Skill of reinforcement

Topic : 9th Std Trigonometry -
Trigonometric ratios

Time : 5 minutes

Date : xxxx

Components of skill

1. Positive verbal reinforcement
2. Positive non verbal reinforcement
3. Negative verbal statement
4. Negative non verbal reinforcement
5. Wrong use of reinforcement
6. Inappropriate use of Reinforcement

Teacher : Good morning Students.

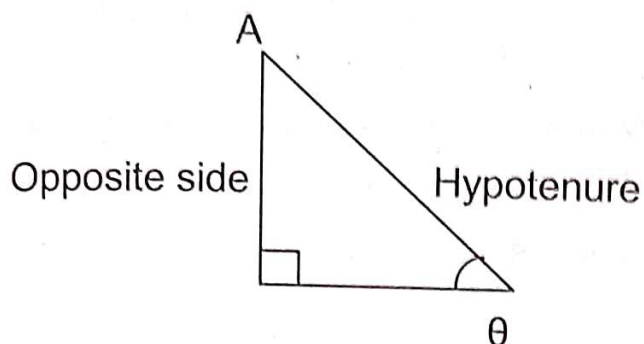
Students : Good morning Sir.

Teacher : What are the different kinds of triangles?

Suganya : Equilateral triangle, scalene triangles, right angled triangle etc.

Teacher : Very good (1) Raju will you come and draw a right angled triangle on the black board?

Raju : Raju draws a right angled triangle and named it ABC



B Adjacent side C

Teacher : Good (1) what is the name of the side AC

Geetha : Longest side of the triangle

Teacher : The longest side of a right angled triangle is called hypotenuse

Teacher : Khader will you tell the statement of Pythagoras theorem?

Khader : In a right angled triangle, the square of hypotenuse is equal to ...

Teacher : 'um' 'um' continue (4)

Khader : the sum of the squares of other two sides

Teacher : Moving towards Khader and patting his shoulder (2)

Teacher : What is the angle C denoted in the figure?

Raju : θ sir

Teacher : Good (1)

Teacher : In a right angled triangle we define

$$\sin \theta = \frac{\text{Opposite side of } \theta}{\text{Hypotenuse}}$$

$$\cos \theta = \frac{\text{Adjacent side of } \theta}{\text{Hypotenuse}}$$

$$\tan \theta = \frac{\text{Opposite side of } \theta}{\text{Adjacent Side of } \theta}$$

Teacher : Mariam, will you tell what is $\sin \theta$, $\cos \theta$ and $\tan \theta$ for the triangle drawn on the black board.

Mariam : $\sin \theta = \frac{AB}{AC}$, $\cos \theta = \frac{BC}{AC}$ and $\tan \theta = \frac{AB}{BC}$

Teacher : Writes on the black board what Mariam told
(2)

Model Micro Lesson - 4

Name of the Teacher trainee : xxxxx

Class : Peer Group

Skill : Skill of closure

Topic : 8th Std - Algebra - Solving first order equations

Time : 5 minutes

Date : xxxx

Components of skill

1. Consolidation of major points
2. Application of present knowledge
3. Linking past knowledge with present knowledge
4. Linking present knowledge with future learning

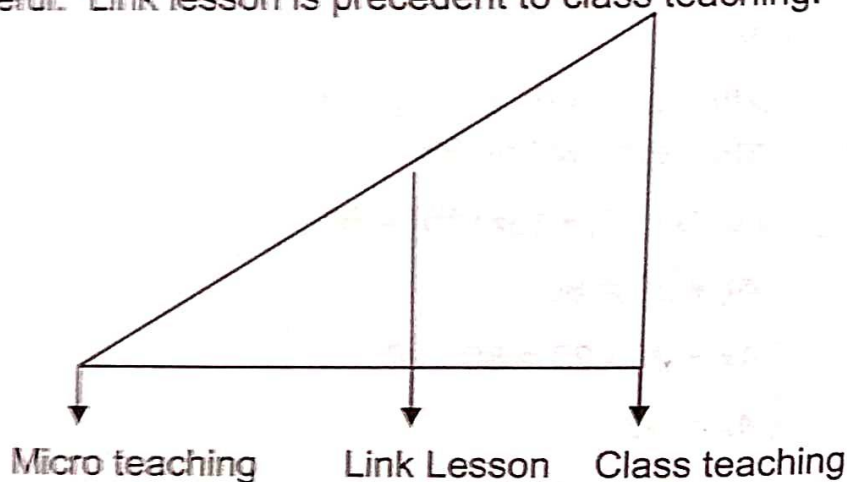
Teacher / Student	Teaching - Learning Activities	Skill components
Teacher	Good morning students	
Students	Good morning Miss	
Teacher	An equation involving one or more variable each with power one is called a Linear equation. $ax+b=0$ is the general form of	Consolidation of major points

Teacher / Student	Teaching - Learning Activities	Skill components
	linear equation is one variable where $a \neq 0$, a, b constants and x is variable. A linear equation in one variable has a unique solution.	
Student	$2x+6=10$, $2x+y=5$	Give any two examples for linear equations.
Teacher	Mary is 3 times older than Nandhini. After ten years the sum of their ages will be 80. Find their present age. Let x be the age of Nandhini, Then the age of Mary is $3x$. After ten years the sum of their ages will be 80. i.e. $(x+10) + (3x+10) = 80$ $4x + 20 = 80$ $4x + 20 - 20 = 80 - 20$ $4x = 60$ $x = 60/4 = 15$ Age of Nandhini = 15 Age of Mary = $3 \times 15 = 45$	Application of present knowledge Linking past knowledge with the present knowledge
Teacher	We have learned how to solve first order equations. In	Linking present knowledge with future learning

Teacher / Student	Teaching - Learning Activities	Skill components
	continuation to this we are going to solve second order equations such as $x^2+5x+6=0$ in the next year	

2.4 Link Lesson

In micro teaching each teaching skill is taken separately and training is given in that skill. All these skills are combined in class teaching. In micro teaching importance is given to teaching skill then the content. In Link Lesson a small content and three or four teaching skills are taken. Importance is given for both content and teaching skills in link lesson. Here teacher trainee is teaching 10 - 15 students in 10 - 15 minutes. In this way if we give few teaching classes to the teacher-trainee to take link lessons and then if we send to class teaching, then it will be more useful. Link lesson is precedent to class teaching.



Micro teaching → Link Lesson → Class teaching

Understanding major Steps in teaching a Link Lesson

While preparing a Link lesson plan, the following steps are followed :

Designing a Link Lesson

The following steps are used in designing a Link Lesson plan.

1. Selecting a content / topic.
2. Setting Lesson objectives.
3. Resource materials needed.
4. Content outline
5. Teaching skills used
6. Instructional Procedures

1. Selecting a Content / topic

The first thing to consider obviously is what you want to teach. This should be based a unit / topic of your syllabus. You also need to be aware of what standard you are developing a topic plan and also the time given for your teaching.

2. Setting Lesson objectives

To make sure your topic plan will teach exactly what you want it to do, you need to develop clear and specific objectives. Note that objectives are the learning outcomes of the lesson. You can certainly have more than one objective for a topic plan.

3. Resource materials needed

Find out what resource materials you are giving to use to explain the topic plan and prepare it well in advance. Be sure the teacher will have materials everything to teach the topic. Power point presentation, Flash Cards, charts or materials are very helpful in teaching your topic.

4. Content outline

Write content key points to be presented.

5. Teaching Skills used

Write the name of the teaching skills used, While teaching a link lesson.

6. Instructional Procedures

- a. Motivation
- b. Presentation
- c. Interaction
- d. Reflection
- e. Summing up

a. Motivation

Students are encourage and motivate to learn new knowledge. The mind of the child should be prepared to receive new knowledge. Preparation may involve.

- 1. Testing previous knowledge to learn new knowledge.
- 2. Integrating the previous knowledge with the new knowledge.
- 3. Arrest attention of the students using new techniques.
- 4. Announcing the aim of the lesson.

b. Presentation

Here teaching takes important place. Students learn new knowledge. For an effective learning outcome, teacher should ensure active student participation with the use of audio-visual aids. Teacher should stimulate thinking by asking number of questions while teaching.

c. Interaction

Interaction means the student teacher activities while teaching is going on in the class. Interaction leads learning. All interaction will take place to attain the objectives.

d. Reflection

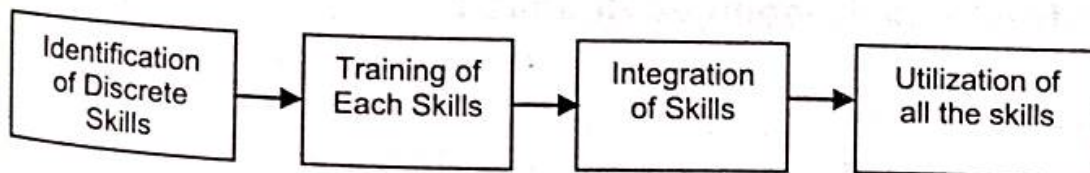
A process of self-observation and self-evaluation is called reflection. By collecting information about what goes on in our classroom and by analysing and evaluating this information is called reflection. We identify and explore our success and failures. This may then lead to changes and improvements in teaching. Reflection can be given by self, peer people or students. Reflection can also be seen with the help of audio and video recording of class teaching.

e. Summing up

After teaching a subject matter in a better way, it is better to close it in the same way. Then only the students feel pleasure and satisfaction. At the end of the lesson, it is necessary to make a quick review what has been taught, and repeating important points.

Practising a Link Lesson

Concept of Link teaching



Link Lesson plan

A Link lesson is a short lesson that provides instruction in one or more skills or concept that students will relate to a larger lesson plan that follow. Link lessons can be used to

teach particular skills, create interest in a topic and generate questions or introduce strategies.

The art of teaching, writing, the Link lesson allows a teacher to convey a content or strategy to students that they will use often. It allows students to gain valuable relevant skills on a regular basis without spending too much of time to teach the same skills. The Link lesson should be short and its duration will be 15 to 20 minutes. Here the teacher introduce the topic, demonstrate skills or concept; guide student practice; discuss the topic; give more examples; and task about what was taught. Link lessons can be taught individually, in a small group or to a whole class.

Link lesson plan format

Name of the Teacher Trainee :

Date :

Standard and section :

Subject : English

Unit :

Topic :

Instructional objectives : 1.

2.

3.

Instructional Resources Required

1.

2.

Content outline

Teaching Skills used

Instructional procedures

Link Lesson - Episode

In the beginning section of this unit we have learnt various skills and their components. This must have significantly contributed to the qualitative improvement of teaching. Now we can integrate all those skills and use them in our teaching. It helps trainee to transfer effectively all the skills.

As episode learnt in the micro teaching on integration of teaching skills is given below for your guidance. Student teachers are asked to study it carefully and plan their practice of integration of teaching skills.

Skills to be integrated

Following skills are integrated and an episode is prepared.

1. Skill of Introducing the lesson
2. Skill of explaining
3. Skill of blackboard usage
4. Skill of questioning
5. Skill of probing questions
6. Skill of stimulus variation
7. Skill of Non-verbal cues
8. Skill of Reinforcement
9. Skill of closure

An episode linking all the above skills is given here under.

Model Link Lesson Plan

Name of the Teacher Trainee	:	xxxx
Date	:	xxxx
Standard and section	:	8 Std. C
Subject	:	Mathematics
Unit	:	Life Mathematics
Topic	:	Compound Interest
Instructional objectives	:	1. Acquires knowledge of compound interest. 2. Tells the difference between simple interest and compound interest. 3. Improve Computation Skill

Instructional Resources Required

Chart depicting formula for simple interest.

Content outline

Motivation - Simple Interest - Example - Meaning of compound interest - Recapitulation - Summing up.

Teaching Skills used

1. Skill of Introducing the lesson
2. Skill of explaining
3. Skill of blackboard usage
4. Skill of questioning

5. Skill of probing questions
6. Skill of stimulus variation
7. Skill of Non-verbal cues
8. Skill of Reinforcement
9. Skill of closure

Instructional procedures

Content	Specification of behavioural activities	Learning Experiences Teacher / Learner activities	Evaluation
Introductory Activities	Motivation	$I = \frac{pnr}{100}$ p = Principal n = Number of years r = rate of interest	What is the formula to find simple interest? What is p, n and r?
Development Activities	Presentation	Today we are going to see compound interest. Calculating interest along with principal and interest is called	What is meant by compound interest?

Content	Specification of behavioural activities	Learning Experiences Teacher / Learner activities	Evaluation
		compound interest	
	Interaction	Yes. Sir. In banks, post office and financial institutions	Will you know where are compound interest used.
Problem :	Vijay deposited Rs. 5,000/- in a financial institution for two years with 10% rate of interest. Calculate the total interest and the total amount he received after two years.		
		Calculating interest for first year using formula. $I = \frac{pnr}{100}$ $\frac{5000 \times 1 \times 10}{100}$ $= 500$ In compound interest second year capital = Capital + First year interest = $5000 + 500$ $= \text{Rs. } 5,500$	What is capital for second year

Content	Specification of behavioural activities	Learning Experiences Teacher / Learner activities	Evaluation
	Inter action	Now we will calculate the interest for second year using formula $I = \frac{pnr}{100}$ $\frac{5500 \times 1 \times 10}{100}$ $= 550$ $500 + 550 = 1050$	What is the total interest Vijay receives after two years?
Concluding Activities		Capital + Total interest $= 5000 + 1050$ $= 6050$	What is the total amount receives by Vijay?
		$I = \frac{5000 \times 2 \times 10}{100}$ $= 1000$	What is the simple interest for Rs. 5000, at 10% for 2 years.
	Reflection	Compound interest	Which is benefit to the customer - simple interest or

Content	Specification of behavioural activities	Learning Experiences Teacher / Learner activities	Evaluation
			compound interest
	Summing up	If the period of time increases above method become lengthy. We will see a formula for compound interest in the next class.	

Link Lesson Episode

Class : 8 Std

Topic - Compound Interest

Teacher / Student	Teaching learning activities	Components
Teacher	Good morning students	
Students	Good morning madam	
Teacher	What is the formula to find simple interest?	Skill of Induction (Previous knowledge)
Student	$I = \frac{pnr}{100}$	
Teacher	What denotes P, n and r	Probing questions. (To increase the

Teacher / Student	Teaching learning activities	Components
		awareness)
Student	$P = \text{Capital}$, $n = \text{time}$, $r = \text{rate of interest}$	
Teacher	Today we are going to see component interest. Calculating interest along with capital and interest is called component interest	Skill of Explaining (Beginning statement)
Teacher	Will you know where are component interest used.	Probing questions (To increase the awareness)
Student	In banks, post offices etc.	
Teacher	Good	Skill of reinforcement (Positive verbal reinforcement)
Teacher	Now we will see how compound interest is calculated. Vijay deposited Rs. 5000/- in an financial institution for two years with 10% rate of interest. Calculate the total interest and the total	Skill of Explaining (Stimulating questions)

Teacher / Student	Teaching learning activities	Components
	amount he received after 2 years.	
Student	We will calculate interest for first year using the formula $I = \frac{pnr}{100}$ $= \frac{5000 \times 1 \times 10}{100} =$ Rs.500	
Teacher	In compound interest, Second year capital = Capital + interest of First year. $= 5000 + 500 = 5,500$ Calculate the Second year interest.	Skill of explaining (Stimulating questions)
Student	$I = \frac{5500 \times 1 \times 10}{100} = 550$	
Teacher	What is the Total interest for two years? What is the amount he will get at the end of Second year.	Skill of explaining (Stimulating questions)
Student	Total interest = Rs. 500 + 550	Skill of explaining (Response to the

Teacher / Student	Teaching learning activities	Components
	$= \text{Rs. } 1,050$ Total amount $= \text{Rs. } 5000 + 1050$ $= \text{Rs. } 6,050$	questions)
Teacher	When the period of time increases, the above method become lengthily. We will see a formula for compound interest in the next class.	Skill of explaining (Concluding statement)

Difference between microteaching, Link lesson and class teaching

Characteristics	Micro teaching	Link lesson	Class teaching
Time	5 - 6 minutes	15 - 20 minutes	40 - 45 minutes
No. of students	5 - 6	10 - 15	40 - 50
Content	Very Small content	Small content	One Lesson or part of it
Teaching skills	Only one skill	Three or four skill	Various skills

REVIEW QUESTIONS

1. Explain in detail the Micro teaching cycle.
2. Explain in detail the skill of set induction with its components and give an episode.
3. Explain the skill of Black board usage with its components.
4. Write the difference between Micro teaching and Macro teaching.
5. Write the major steps in teaching a Link Lesson.
6. Write a Link lesson episode for 7th Standard Mathematics.